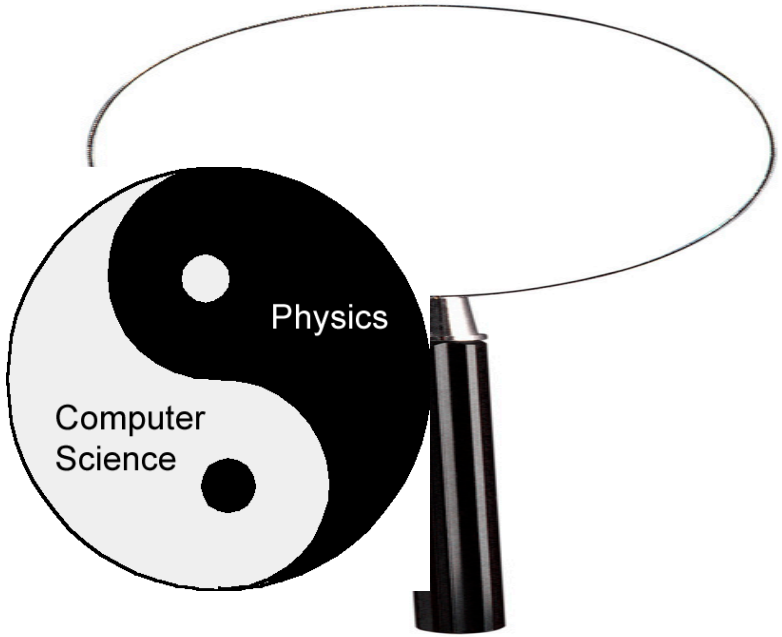
# Physics and Computer Science Through the Lens of Quantum Computing

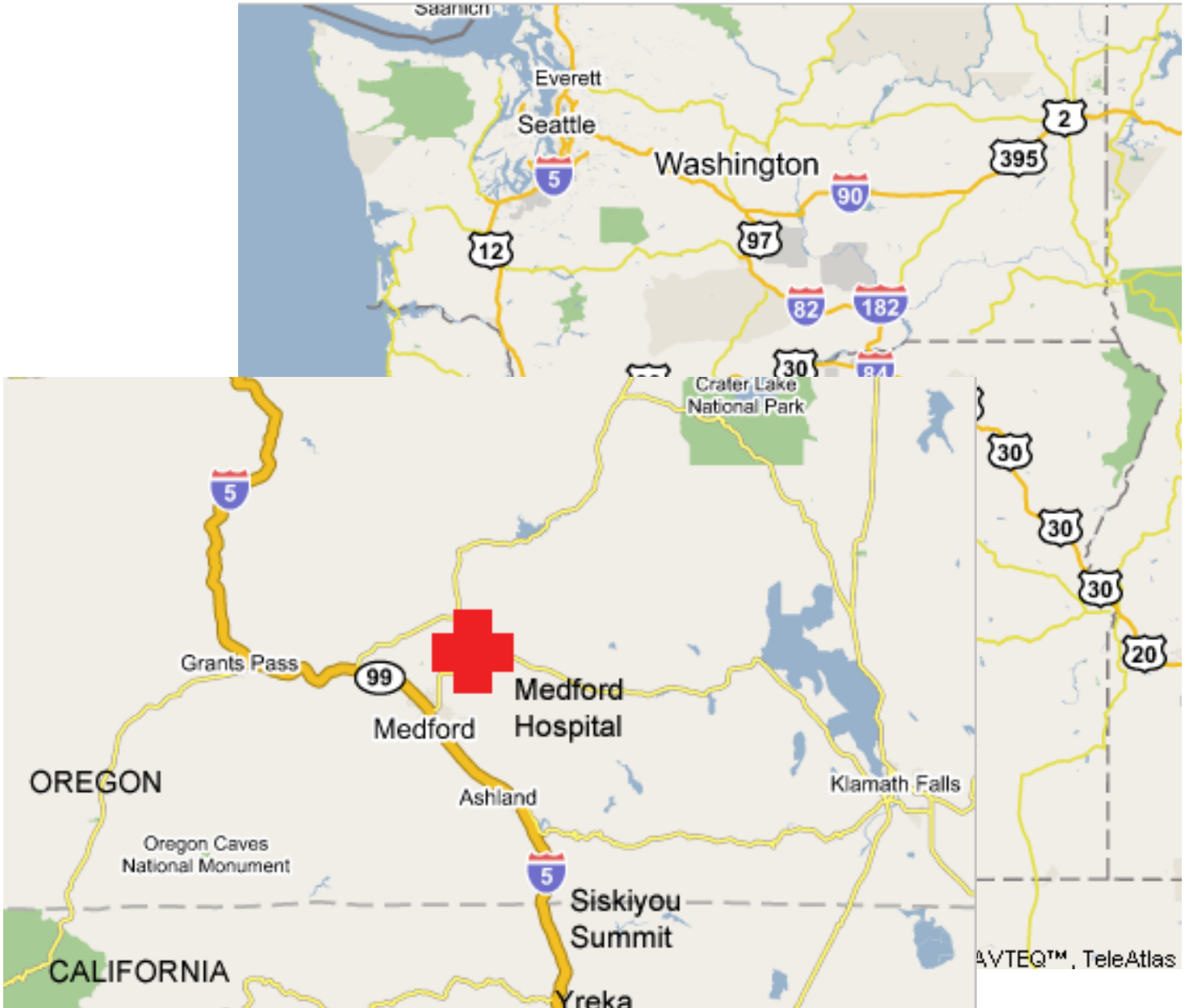
**Dave Bacon**

Department
Univers

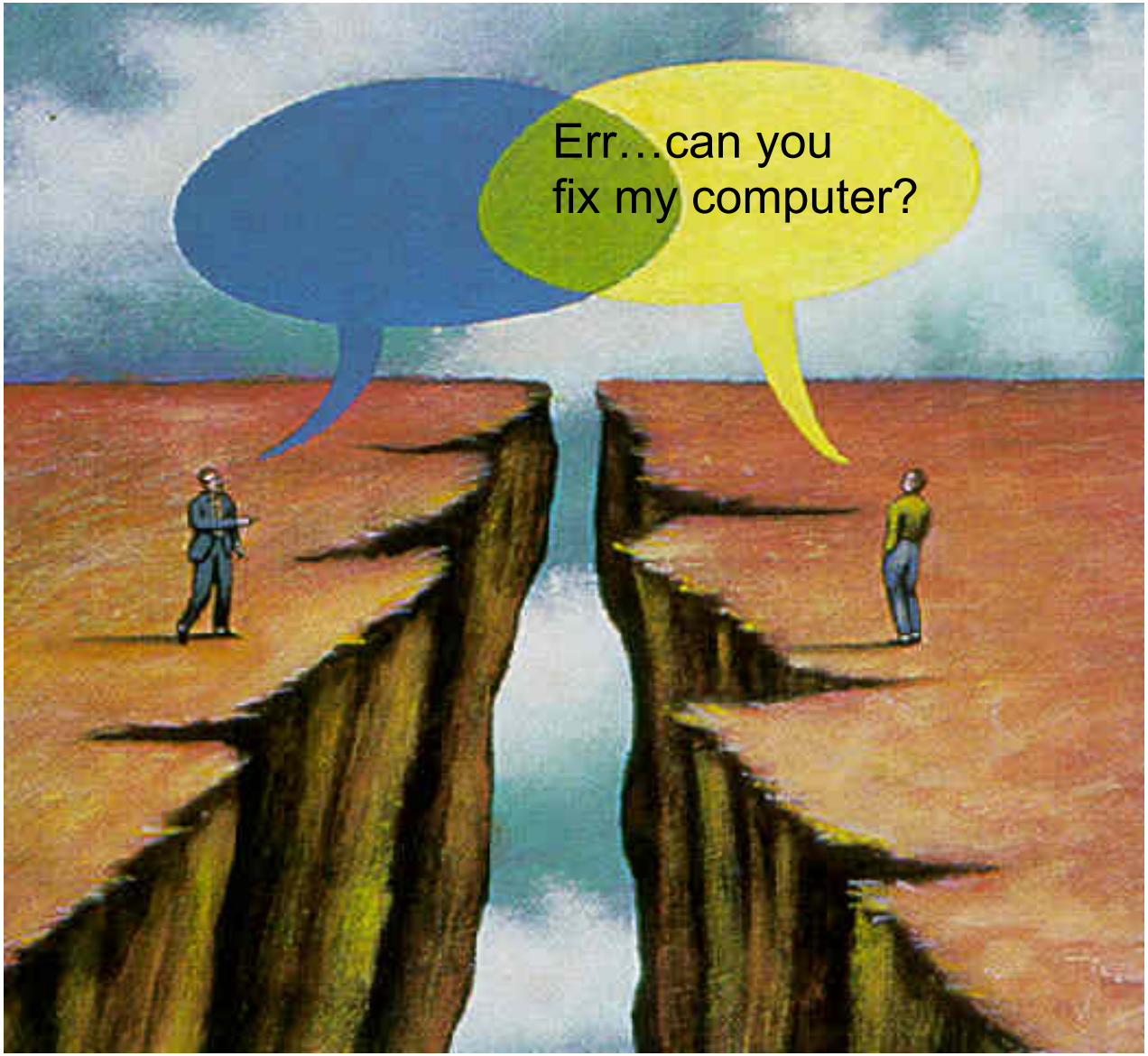




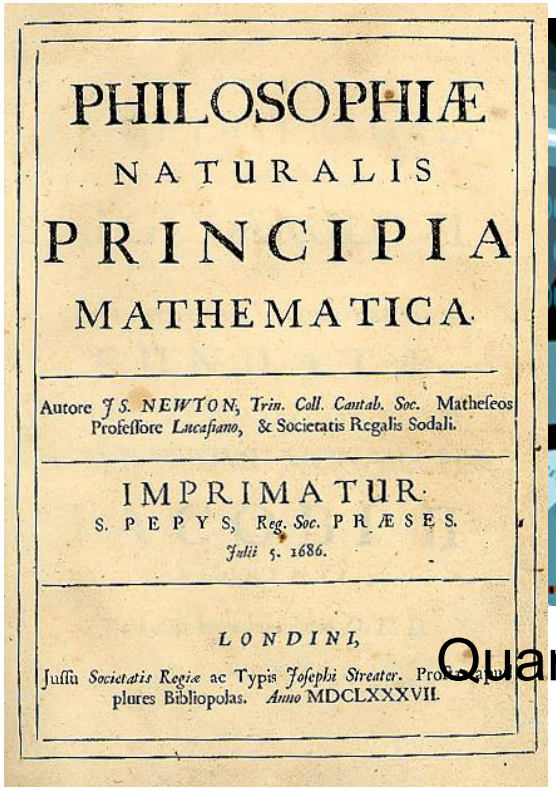

# All Apologies



# Physics and Computer Science



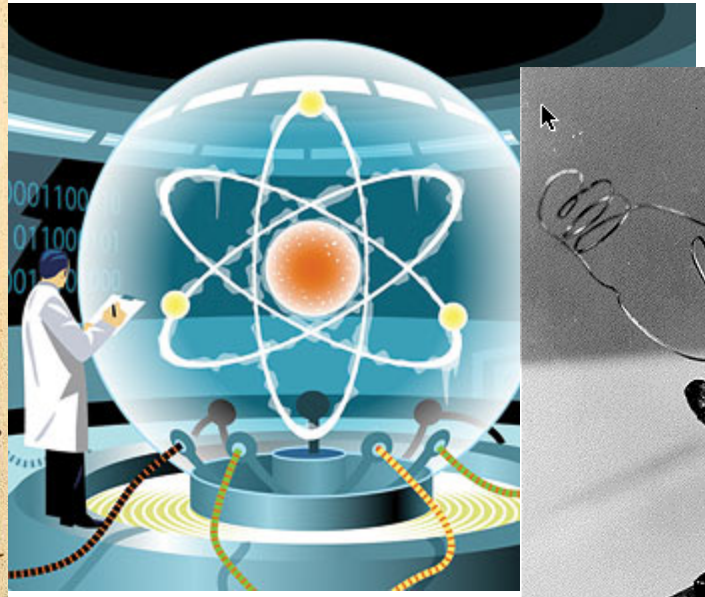
# What Gets Me Out of Bed

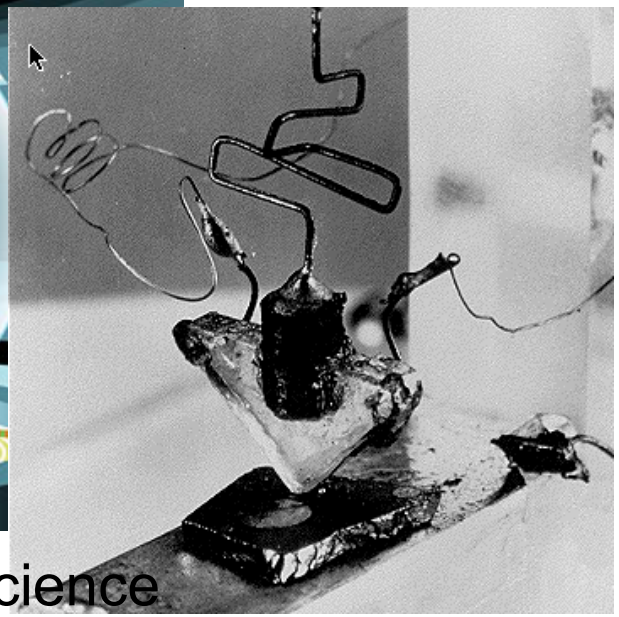

PHILOSOPHIÆ
NATURALIS
PRINCIPIA
MATHEMATICA

Autore *J S. NEWTON, Trin. Coll. Cantab. Soc.* Matheseos Professore *Lucasiano*, & Societatis Regalis Sodali.

IMPRIMATUR
S. PEPYS, *Reg. Soc.* PRÆSES.
*Julii* 5. 1686.

*LONDINI,*
Jussu *Societatis Regiæ* ac Typis *Josephi Streater.* Prostat apud plures Bibliopolas. *Anno* MDCLXXXVII.



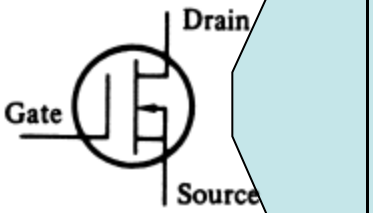


Quantum Information Science

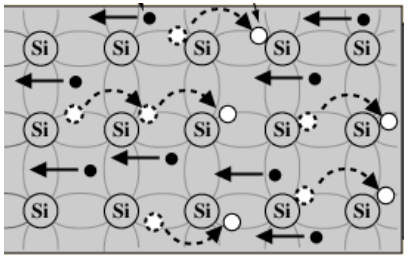
Intellectual Merit

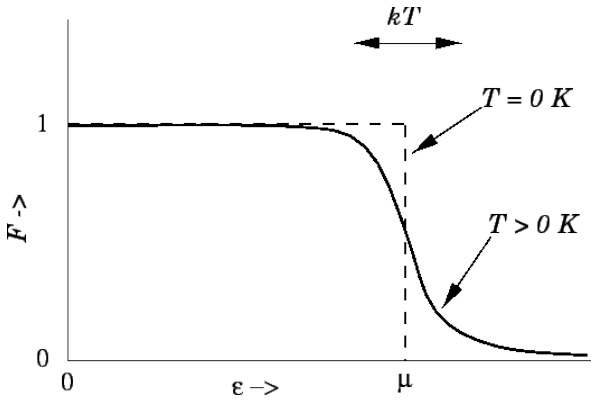
Practical Consequences

# Behind the Bit

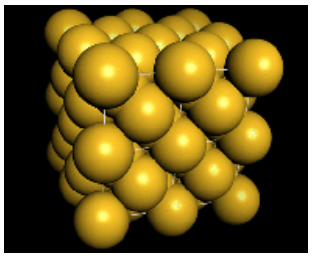


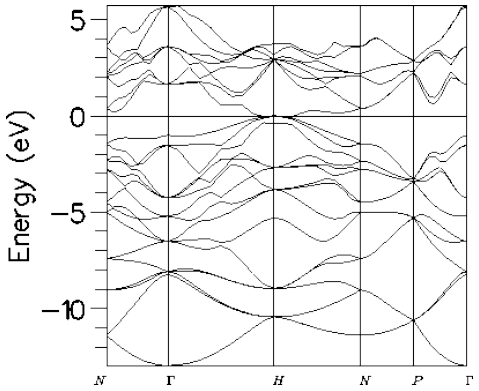

bits




noisy bits






quantum bit

# Quantum Computer?

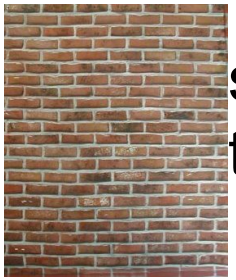


A Famous Bongo Player

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It's not a Turing machine but a machine of a different kind"

- Richard Feynman (1981)





simulating quantum theory on a computer

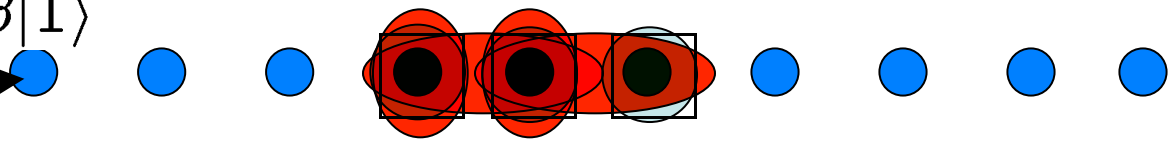
# Bit, Noisy bit, and Qubit

| | bit | noisy bit | quantum bit |
|---|---|---|---|
| Configurations: | 0 1 | 0 1 | 0 1 |
| Description: | $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ | $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$ | $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ $\alpha, \beta \in \mathbb{C}$ |
| Observation: | 0 certainty | 0 *p* percent<br>1 *1-p* percent | 0 $\lvert\alpha\rvert^2$ercent<br>1 $\lvert\beta\rvert^2$ercent |
| Evolution: | $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ | $\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$ | $\begin{bmatrix} u & v \\ w & x \end{bmatrix}$ |
| | deterministic | stochastic | unitary |

# The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

qubits



$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement

0

1

1

# Quantum Computing

**FACTORING:** Given an *n* digit positive integer *N*, find its factors.

188198812920607963838697239461650439807163563379417382700763356422988859715234654853190606065047430453173880113033967161969232120573403187955065699622130516875930765025705 9

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Conventional Computers: $O(\exp(n^{1/3}))$

Quantum Computer: $O(n^2 \log n)$ [Peter Shor 94]



Public Key Cryptosystems

RSA
Diffie-Hellman
Elliptic Curves

**BROKEN**



# Over the Top



“Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world’s most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world.  Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security.” - Simon Singh

# Quantum Algorithms




Quantum algorithms > Classical algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- Simulation of quantum systems

Applications to chemistry, biology, material science….

# Quantum Computing?

What merit? (beyond new algorithms which scare the NSA)

**What** gives quantum computers their **power** over classical computers

**How** building a quantum computer will result in a **new phase of matter**

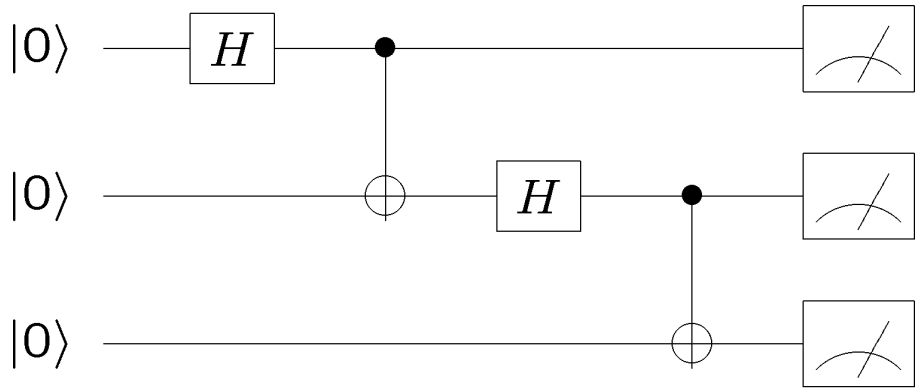
# Quantum Computer = Parallel Computer?

n qubits = superposition over 2n states

$$|\psi\rangle = \sum_{x=0}^{2^n-1} \alpha_x |x\rangle$$

"massively parallel computer"?

**NO!**




Measurement results in only one outcome

Compare:

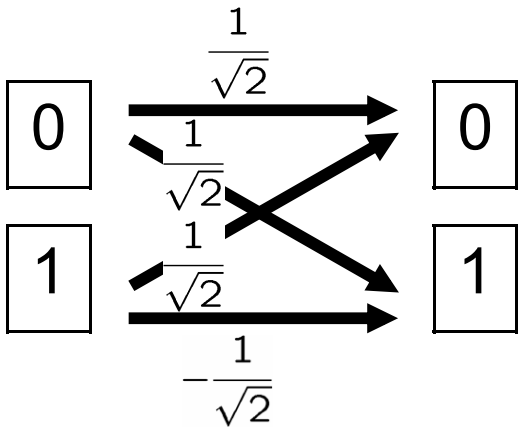
n noisy bits = convex sum of 2n states

$$\rho = \sum_x p_x |x\rangle\langle x|$$

# Interference?








$$:= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

qubit input

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

measure

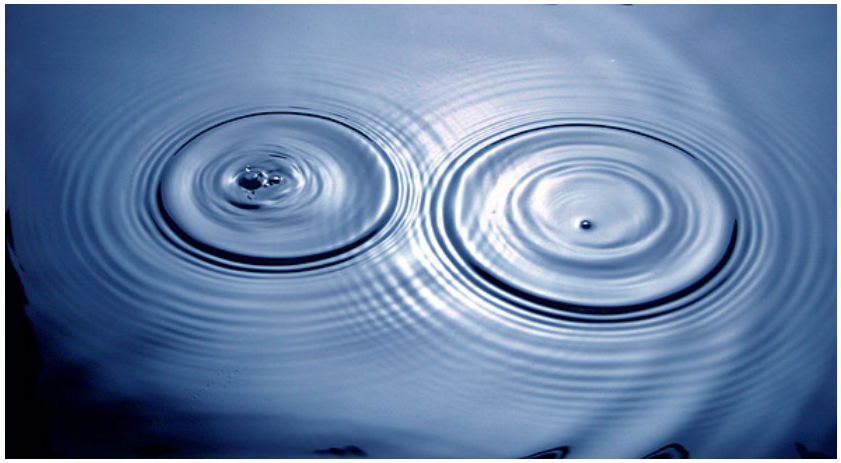
| | | |
|---|---|---|
| 0 | 50% | 100% |
| 1 | 50% | 0% |

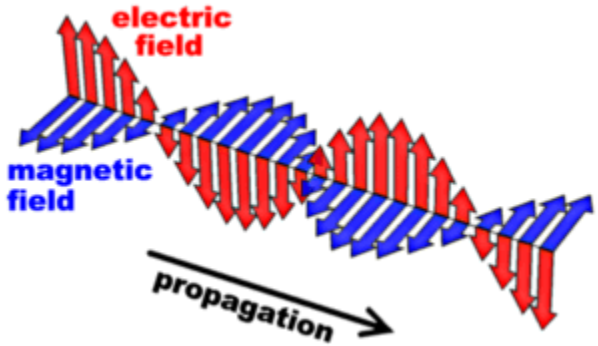
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

measure

| | | |
|---|---|---|
| 0 | 50% | 0% |
| 1 | 50% | 100% |

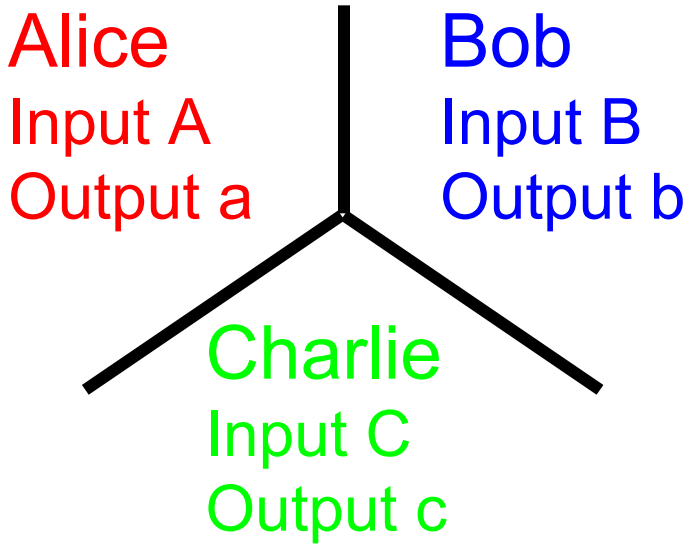
# Not Interference






Interference is not only quantum mechanical

# GHZ/Mermin Game




Input: X or Y, Output: +1 or -1

Winning Payoffs:

| A | B | C | abc |
|---|---|---|---|
| X | X | X | +1 |
| Y | Y | X | -1 |
| Y | X | Y | -1 |
| X | Y | Y | -1 |

deterministic local strategies, given by aX aY bX bY cX cY

aY bY cX = -1

aY bX cY = -1

× aX bY cY = -1

---

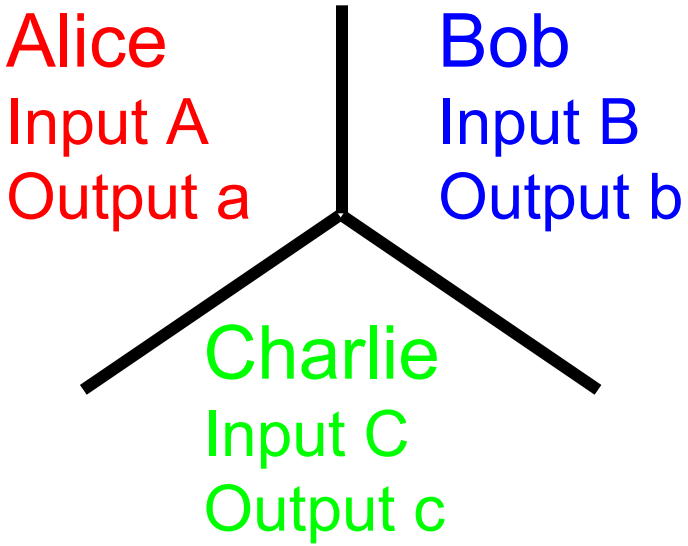
aX bX cX = -1

→ not possible to always win

→ even with <u>prior</u> classical communication

convex sum of deterministic local stratgies

# GHZ/Mermin Game




Input: X or Y, Output: +1 or -1

Winning Payoffs:

| A | B | C | abc |
|---|---|---|---|
| X | X | X | +1 |
| Y | Y | X | -1 |
| Y | X | Y | -1 |
| X | Y | Y | -1 |

Quantum communication before game is played:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

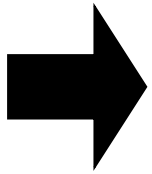
Each party measure along X or Y direction

$\langle GHZ|XXX|GHZ\rangle = +1$

$\langle GHZ|YYX|GHZ\rangle = -1$

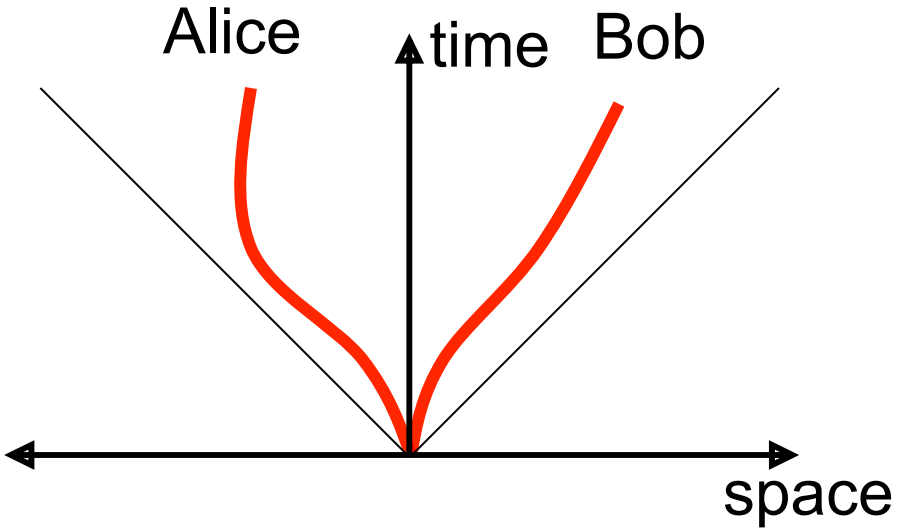
$\langle GHZ|YXY|GHZ\rangle = -1$

$\langle GHZ|XYY|GHZ\rangle = -1$



possible to always win!

# Bell's Theorem






John Bell

Separable: $|\psi_1\rangle \otimes |\psi_2\rangle$

Entangled: $\neq |\psi_1\rangle \otimes |\psi_2\rangle$

Measurements on space-like separated entangled quantum systems cannot be explained by local classical communication.

# Minimizing the Yakking



ALICE

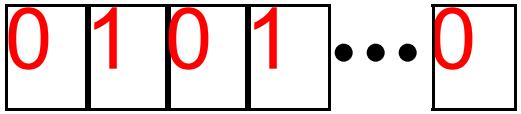
a

Feq(a,b)= 1 if a=b
0 if a≠b

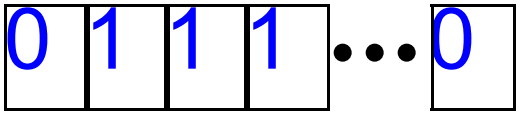


BOB

b




n bits




n bits

How many bits do Alice and Bob need to exchange (in the worst case) to compute F(a,b)?

*With no error:* n+1 bits

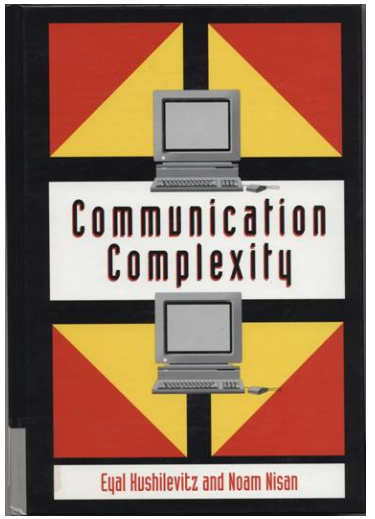
*With one-sided error ε:* O(log(n/ε)) bits

# Communication Complexity



Classical communication costs of distributed classical computations

[Yao, STOC 79]

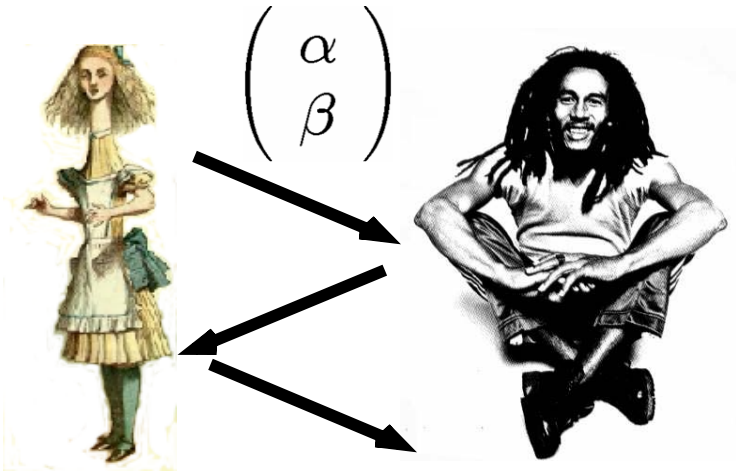



[Kushilevitz and Nisan 96]



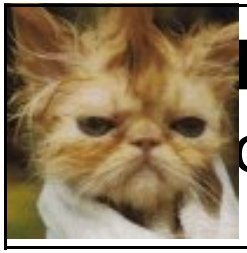
Quantum communication costs of distributed classical computations

[Yao, FOCS 93]




exchange qubits

# Quantum Communication Complexity



**Holevo's Theoerem:** sending n classical bits over a quantum channel requires n qubits



[Raz, STOC 99]

There exist functions *F(a,b)* which require $\Omega(m1/4/\log(m))$ bits of communication to solve (with error) but can be solved using only $O(\log(m))$ qubits of communication.
(*m* is the length of *a* and *b*)



**Exponential** benefit of quantum over classical.

# Teleportation (Demystified)



Alice wants to send $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ to Bob

$\begin{bmatrix} p \\ 1-p \end{bmatrix}$

1 bit





A

$\begin{bmatrix} p \\ 1-p \end{bmatrix}$

1 bit

AB

$\begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$ 00 01 10 11

B



$\begin{bmatrix} p \\ 1-p \end{bmatrix}$

If Alice's bit is 1, then flip A

If received bit is 1, then flip B

# Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob



$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ⟶ 1 qubit



X

Y

2 bits ⟶





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ○ $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ ○ $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

Conditional on received bits, Bob performs manipulation of his half of entangled system

# Rehash



[Raz, STOC 99]

There exist functions *F(a,b)* which require *Ω(m1/4/log(m))* bits of communication to solve (with error) but can be solved using only *O(log(m))* bits and *O(log(m))* ebits.
(*m* is the length of *a* and *b*)

1 ebit $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$

1. Entanglement can be used to gain exponential reduction in communication resources

2. Entanglement is “like” classically correlated bits…but it’s different…

**Preliminary thesis:** entanglement is involved in quantum benefits

# Quantifying Entanglement

Pure state bipartite entanglement

$$\left[\sum_{i=0}^{r-1}\sqrt{p_i}|\phi_i\rangle\otimes|\varphi_i\rangle\right]^{\otimes m} \longleftrightarrow \left[\frac{1}{\sqrt{2}}(|0\rangle\otimes|0\rangle+|1\rangle\otimes|1\rangle)\right]^{\otimes n}$$

$$\frac{n}{m}=S=-\sum_{i=0}^{r-1}p_i\log_2 p_i$$

Entropy of one-half of mixed state is measure of entanglement

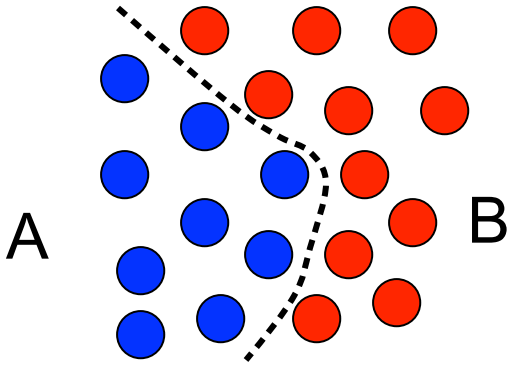
Schmidt decomposition: *r* is the Schmidt rank

Schmidt rank is upper bound on entropy: $S \leq r$

Crucial insight: *r* numbers to describe entangled state

# “Small” Entanglement

*n* qubits




$$\left[\sum_{i=0}^{r-1}\sqrt{p_i}|\phi_i\rangle\otimes|\varphi_i\rangle\right]^{\otimes m}$$

$$\chi = \max_A r$$

Theorem [Vidal 03]: If $\chi$ is *O(poly(n))* then there is an efficient description of the state. If $\chi$ remains small throughout a quantum computation then there is a *O(poly(n))* classical simulation of the quantum computation.

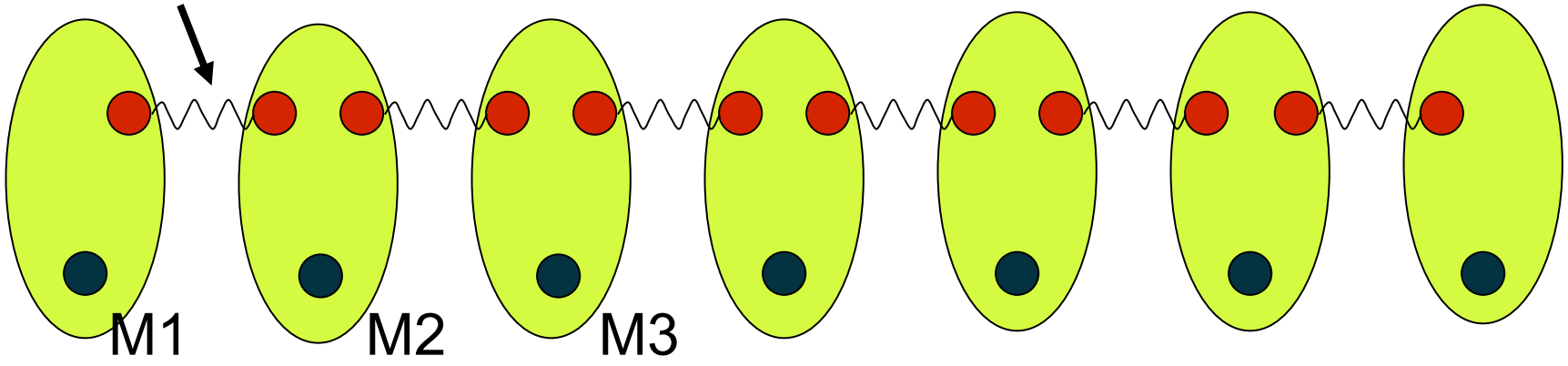
“Small” Entanglement = Efficiently Classical Simulation

# Matrix Product States

"Small" Entanglement = Efficiently classical simulation

? Better classical algorithms for simulating real world quantum systems?

$$\frac{1}{\sqrt{D}} \sum_{i=0}^{D-1} |\phi_i\rangle \otimes |\varphi_i\rangle$$

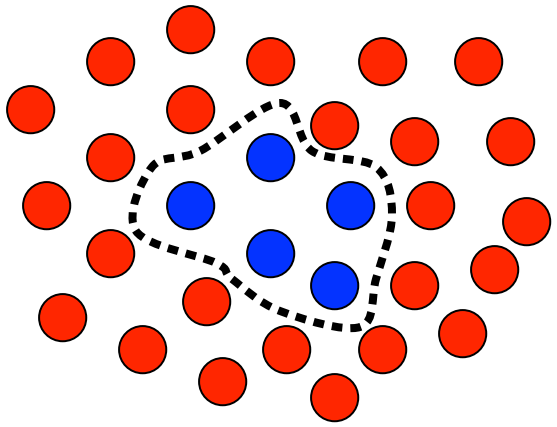



Matrix Product States: (Each Mi requires *D2s* numbers)

Schmidt rank bounded by *D* ⇒ D small, efficient simulation

Density Matrix Renormalization Group [White]

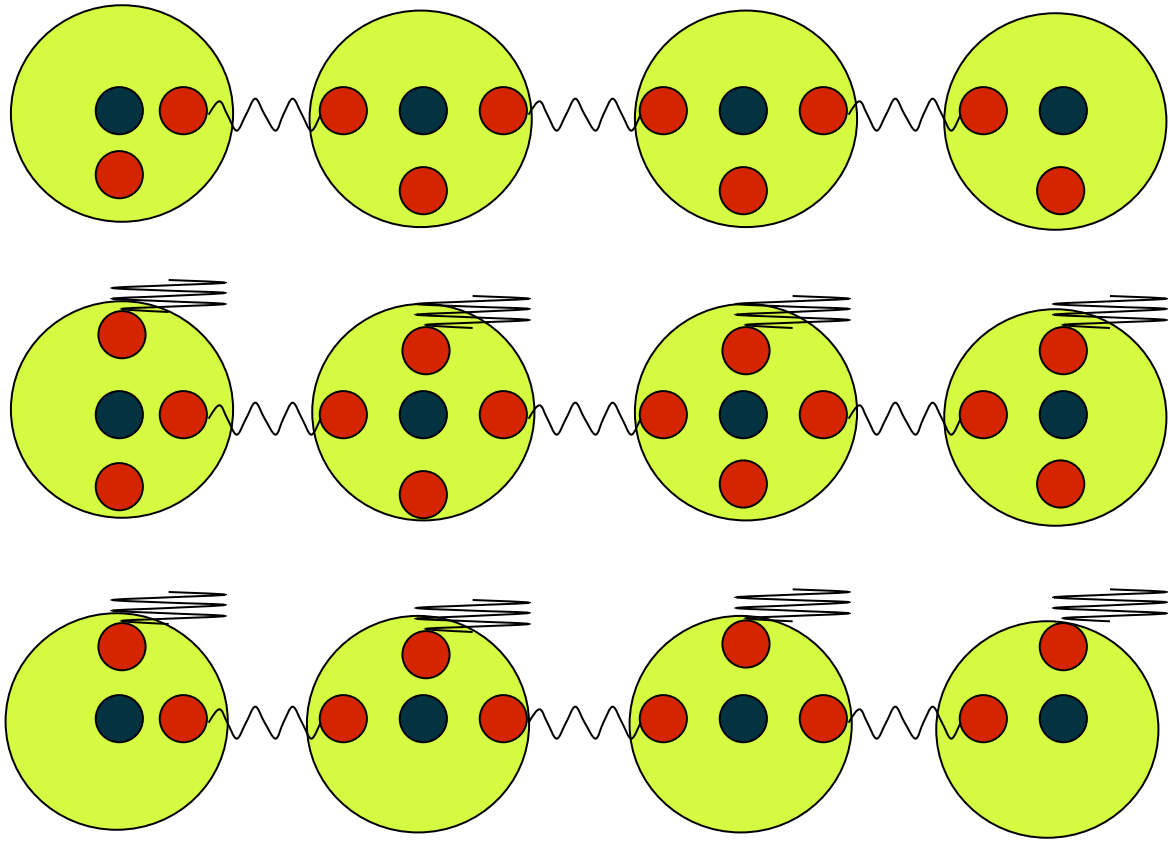
# Perimeter “Law”



In many quantum systems, the low energy sector has an entropy of a region which grows like the perimeter of the region

“Small” entanglement

# DMRGish Algorithm in 2D



Design states which respect this scaling!

Efficient algorithms for simulating low energy sector of high (spatial) dimensional systems. [Verstraete, Cirac 04]

# Quantum Computing?

**What** gives quantum computers their **power** over classical computers

New classical algorithms for use by physicists!

Inverting promise of quantum computing!

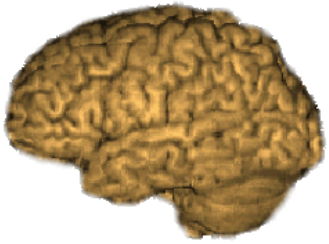
# Quantum Computing?

**How** building a quantum computer will result in a **new phase of matter**

# Two Views of Information

**Information is "fungible"** adj. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.







Al Gore

George Bush

Ralph Nader

Pat Buchanan

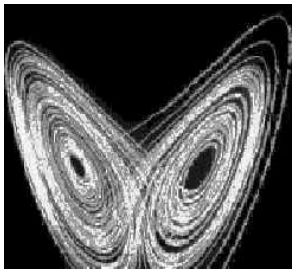


**No Information without Physical Representation!**



Rolf Landauer

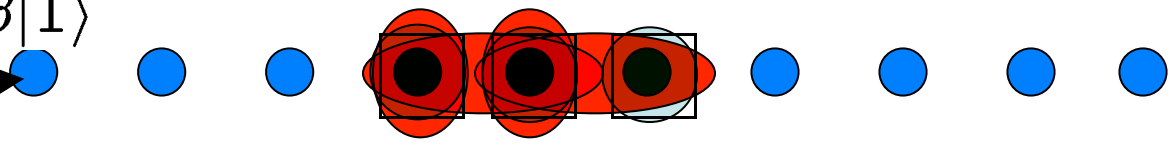
Not all physical systems are robust information processing machines



# The Ideal Quantum Computer

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

qubits



$\frac{1}{2}(|000\rangle + |011\rangle + |100\rangle - |111\rangle)$

Quantum algorithm

1. Initialize
2. Series of unitary gates
3. Measurement

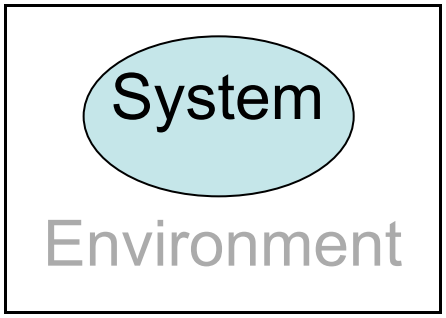
0

1

1

# Obstructions

**DECOHERENCE**




Total Hamiltonian

$$H = H_S \otimes I + I \otimes H_E + H_I$$

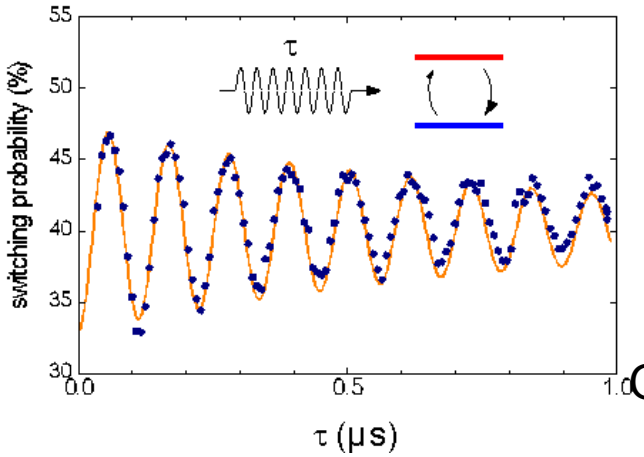
system environment interaction

open quantum systems can become *entangled* with their environment

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

coherent superposition incoherent sum

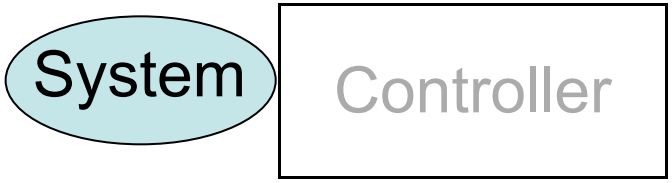
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$



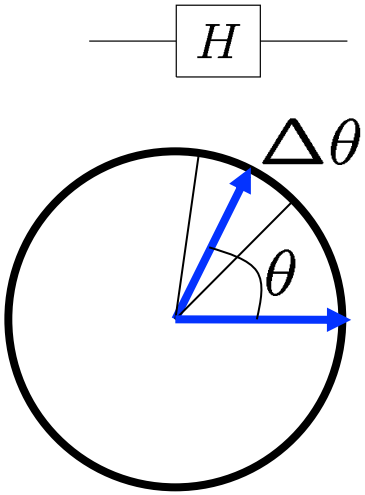

Quantronium (CEA, France)

# More Obstructions

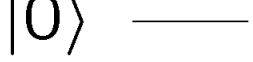
**FAULTY CONTROL**




Imprecise or faulty unitary gates

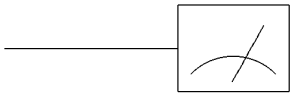



Imperfect state preparation




$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

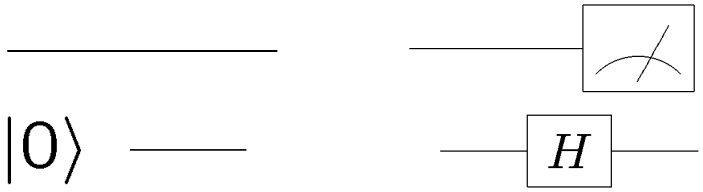
Imperfect measurements



$$\alpha|0\rangle + \beta|1\rangle \begin{cases} \rightarrow Pr(0) = |\alpha|^2 - \epsilon \\ \rightarrow Pr(1) = |\beta|^2 + \epsilon \end{cases}$$

# This is the End, My Only Friend, the End

Everything, (quantum) noisy, all the time.




# Is a Quantum Computer Even a Computer?

A skeptics checklist:



Laughlin

1. quantum data is analog
2. quantum data likes to become classical data
3. quantum evolution is analog
4. quantum theory is wrong

# Quantum Data is Analog

n qubit quantum state: $|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle$



**Objection:**

Superposition of an exponential number of states.

Each amplitude is a complex number…i.e. analog data

How do you control 2n complex numbers?

In the real world, nothing can be done with infinite precision

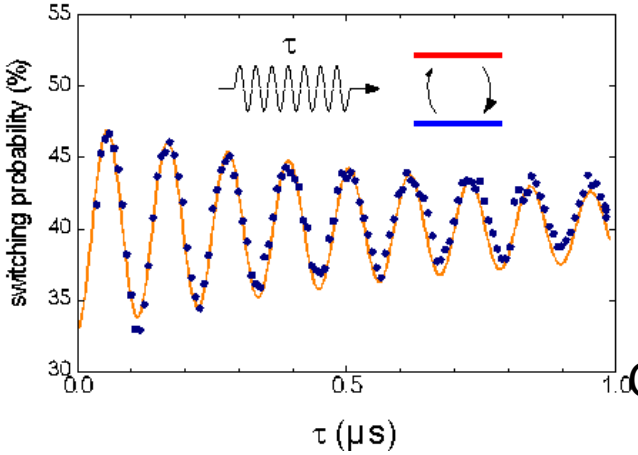
# Quantum data likes to become classical data

coherent superposition

incoherent sum

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2|0\rangle_S\langle 0|_S + |\beta|^2|1\rangle_S\langle 1|_S$$

Decoherence is a complex process which is fundamentally responsible for there being classical information




Quantronium (CEA, France)

Decoherence is irreverislbe

# Quantum Evolution is Analog

Quantum evolution: $U(t) = \exp(-iHt)$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

A unitary “gate” also has a bunch of complex numbers

Each amplitude is a complex number…i.e. analog data

In the real world, nothing can be done with infinite precision

# Quantum Theory is Wrong

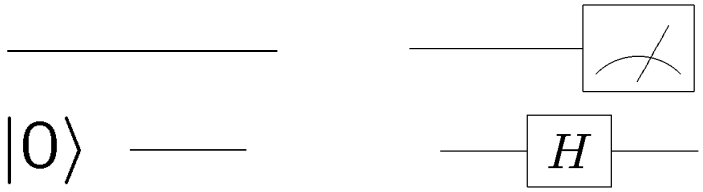
Quantum theory has only ever been observed for small (micro to meso) systems.

It will break down for large systems



# This is the End, My Only Friend, the End

Everything, (quantum) noisy, all the time.




# Is a Quantum Computer Even a Computer?

A skeptics checklist:

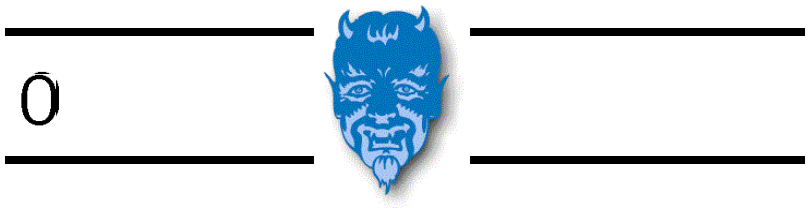


Laughlin

1. quantum data is analog
2. quantum data likes to become classical data
3. quantum evolution is analog
4. quantum theory is wrong

# Why Is Classical Computation Possible?

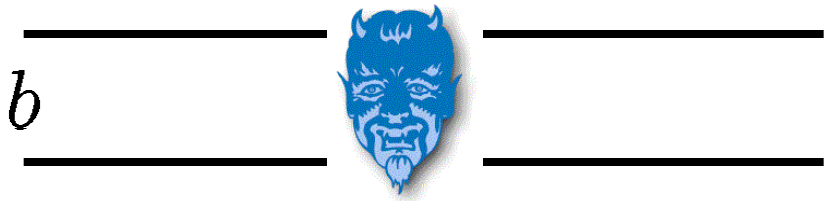
# Classical Solutions




Probability of bit flip = *p*

Solution: redundancy

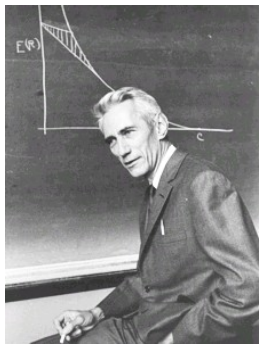
$0 \rightarrow 000$
$1 \rightarrow 111$




$$\left.\begin{matrix} 000 \\ 001 \\ 010 \\ 100 \end{matrix}\right\} \rightarrow 0$$

$$\left.\begin{matrix} 111 \\ 110 \\ 101 \\ 011 \end{matrix}\right\} \rightarrow 1$$

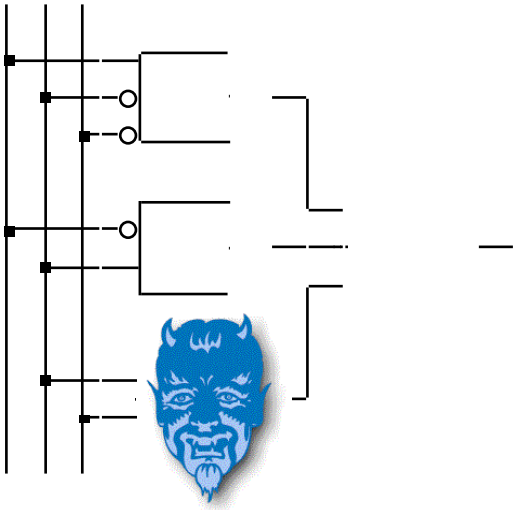
Probability of failure =*3(1-p)p2+p3*
=*3p2-2p3*
<*p* if *p*<*1/2*



Use redundancy to protect information against noise

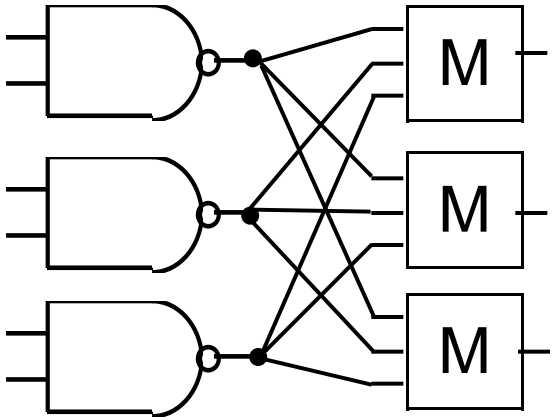
Claude Shannon 1948
“A mathematical theory of communication”

# More Classical Solutions



Classical gates fail with probability p

Solution: error correcting codes + fault tolerant construction



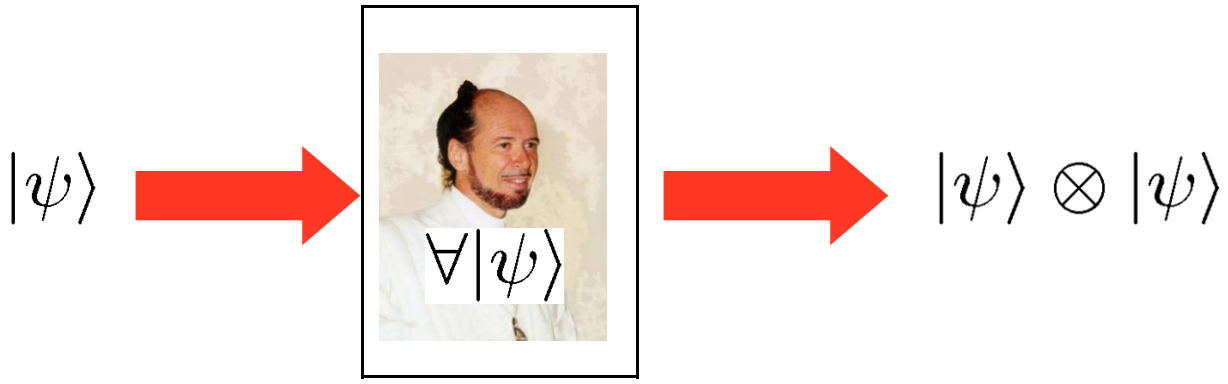



John von Neumann, 1956

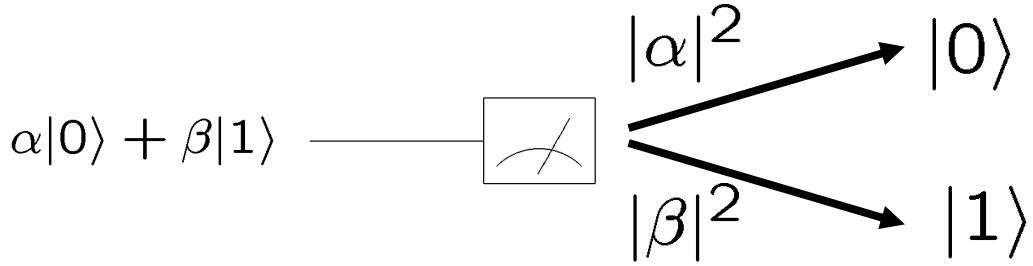
“Probabilistic logics and the synthesis of reliable organisms from unreliable components”

# Quantum Problems?

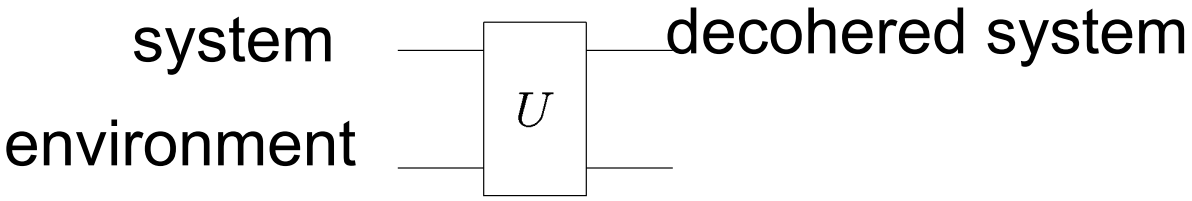
No cloning of quantum information




How to perform error correction without measuring?

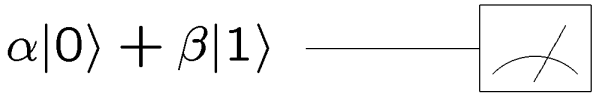



Quantum errors seem “analog”

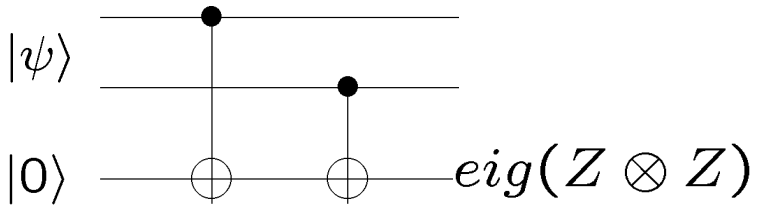


# Quantum Observations

How to perform error correction without measuring?




Observation: not all measurements destroy quantum information




$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

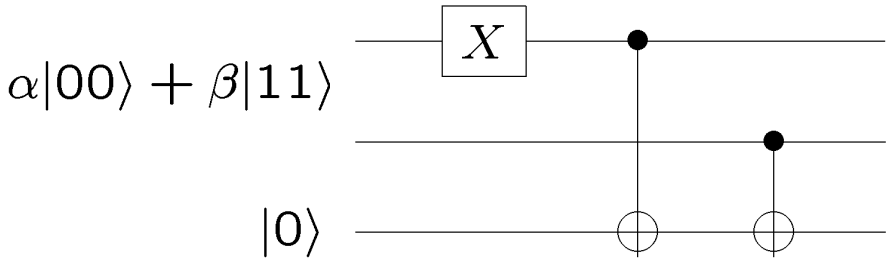
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

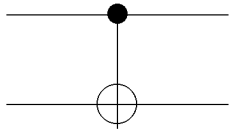
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Information encoded into a subspace undisturbed

$$\alpha|00\rangle + \beta|11\rangle \rightarrow \alpha|00\rangle + \beta|11\rangle \qquad eig(Z \otimes Z) = +1$$

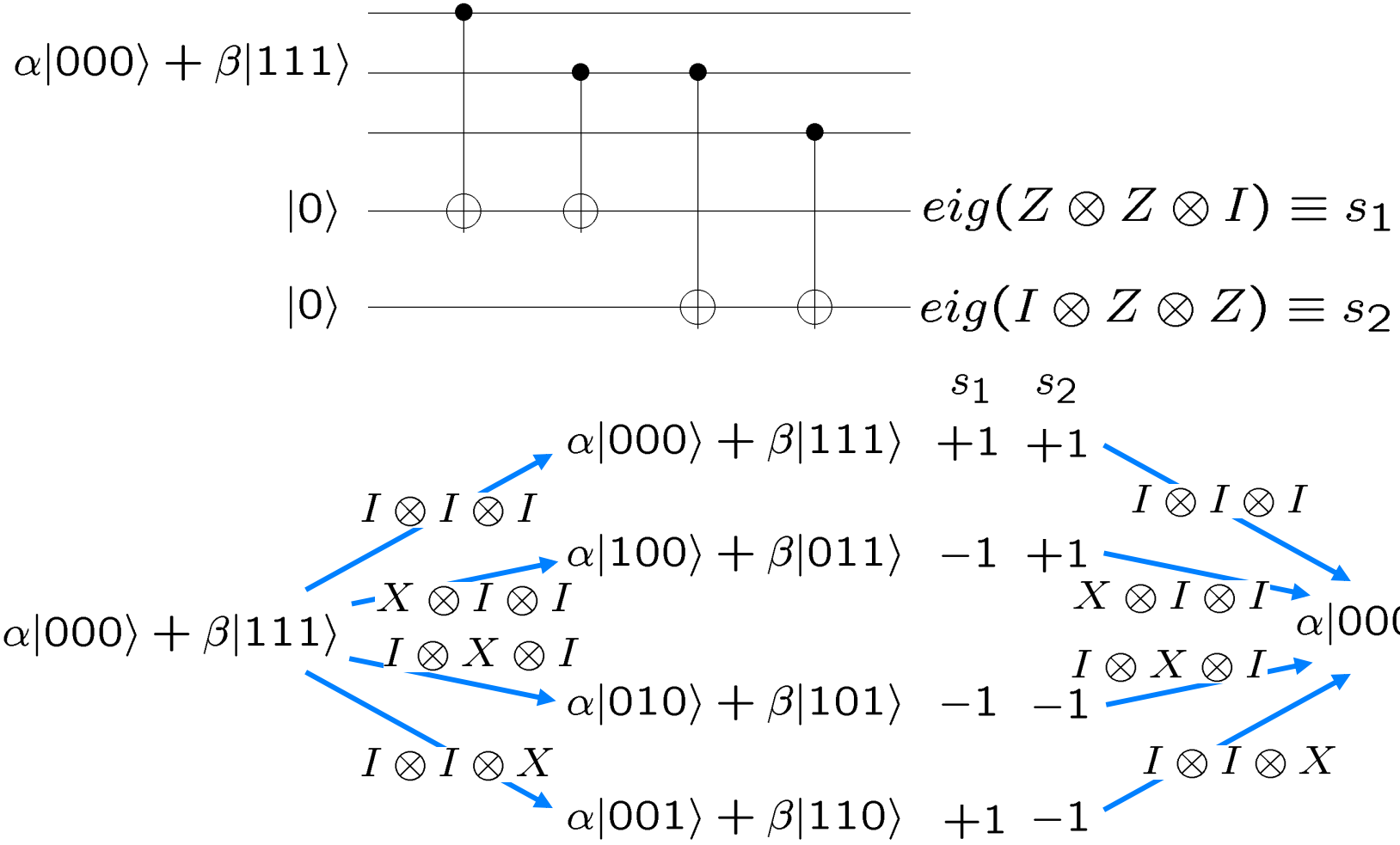
Can detect “leaving” of subspace



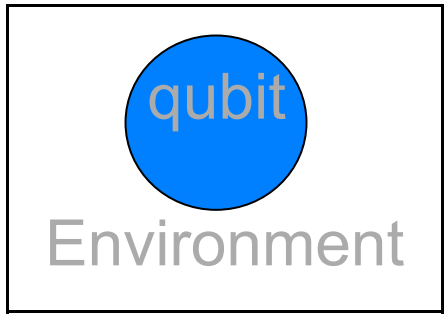
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

# Quantum Observations

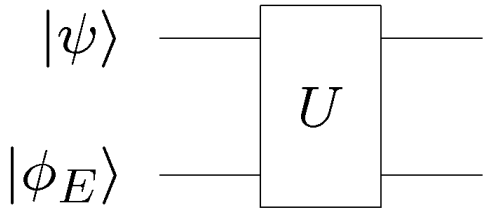



By encoding into subspaces we can correct single bit flips

# Quantum Problems?




$$H = H_S \otimes I + I \otimes H_E + H_I$$




$$|\psi\rangle\otimes|E_I\rangle+(X|\psi\rangle)\otimes|E_X\rangle+(Y|\psi\rangle)\otimes|E_Y\rangle+(Z|\psi\rangle)\otimes|E_Z\rangle$$

digitized effects of decoherence

Think: amplitude for no error, *X* error, *Y* error, *Z* error

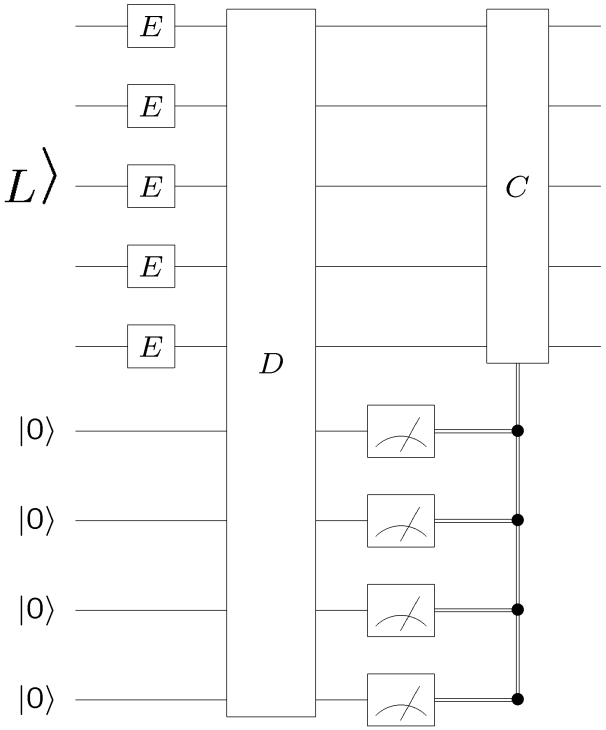
# Quantum Error Correcting Codes

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I \qquad [s_i, s_j] = 0$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

Encode qubit into 2 dimensional subspace with all +1 eigenvalues

Single qubit errors (i.e. $I \otimes X \otimes I \otimes I \otimes I, I \otimes I \otimes Y \otimes I \otimes I$) all lead to a unique subspace labeled by *si* eigenvalues

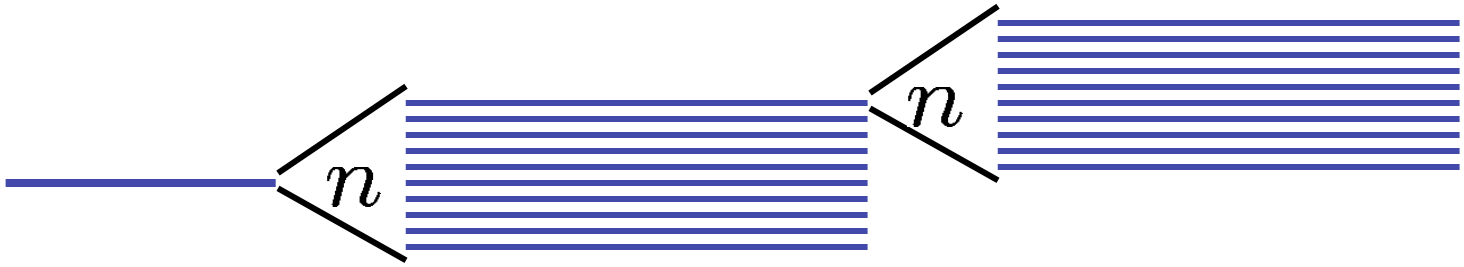
$\alpha|0_L\rangle + \beta|1_L\rangle$




$\approx \alpha|0_L\rangle + \beta|1_L\rangle$

"bare" error rate = *p*
error rate on encoded = *O(p2)*

# Perfection Through Concatenation




failure probability

*p* *cp2* *c(cp2)2*

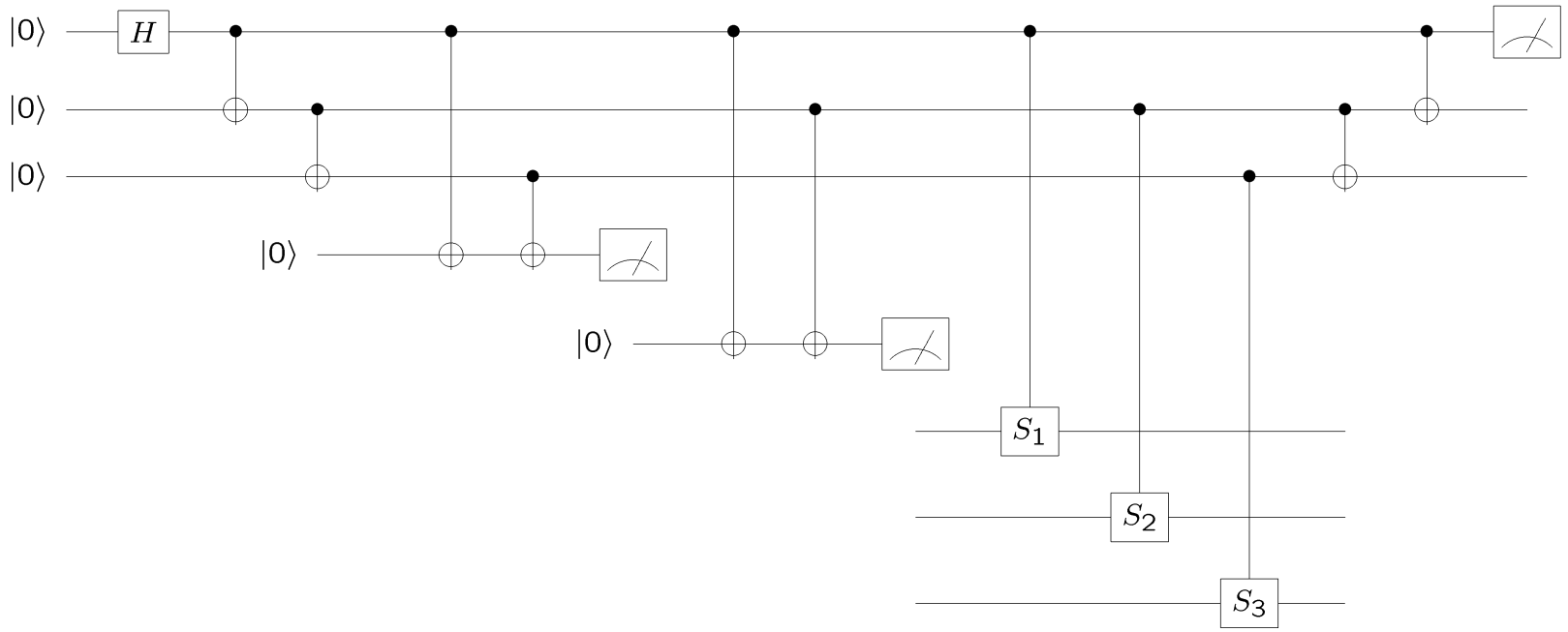
At kth level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c} \qquad \frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k} \qquad p_{th} = \frac{1}{c}$$

and uses $N_{qubits} = n^k$ qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}}\left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log\frac{1}{p_{fail}}\right)$$

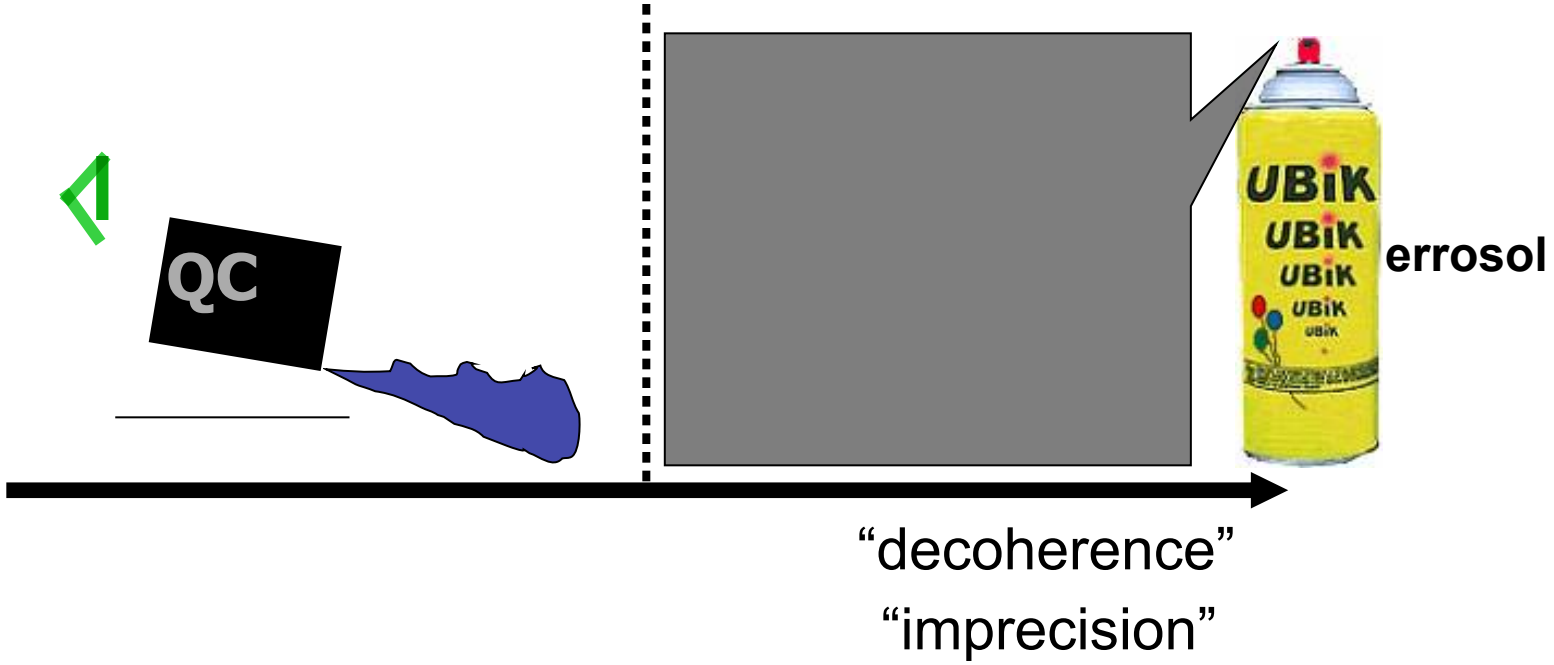
# Fault-Tolerant Quantum Computation




Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, measurements) can fail.

**Quantum error correction + Fault-tolerant procedures**

# Quantum Threshold Theorem




**Threshold Theorem for Fault-Tolerant Quantum Computation**

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Kitaev, Gottesman, Preskill, Shor, + others]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

# Lessons

1. quantum data is analog

| Degrees of freedom | Description of system |
| --- | --- |
| analog | deterministic |
| digital | probabilistic |
| | quantum |

2. quantum data likes to become classical data

quantum information can be encoded
encodings can reduce effective error rate
quantum noise can be “digitized”

3. quantum evolution is analog

encoded quantum information can be computed on
fault-tolerant constructions can be error corrected

4. quantum theory is wrong

maybe, but then…what a discovery!!!

# Threshold(s) Theorem(s)

Threshold theorems come in many different forms….

provable bounds, Monte Carlo estimates, …

different shapes…

assumptions of error models
assumptions of physical devices
assumptions of architectures

yielding threshold values for relative error rates from
10-9 to 10-2

# History

1994-5 Shor, Steane invent quantum error correction

Shor invents FT protocols

1996

Aharonov, Ben-Or
FT Proof

Knill, Laflamme, Zurek
FT Estimate

Zalka
Threshold estimate
(10-4)

Kiteav
Topological QC
FT Proof

1997-2003 Everyone admires how cool the threshold theorem is

2004-NOW

Renaissance

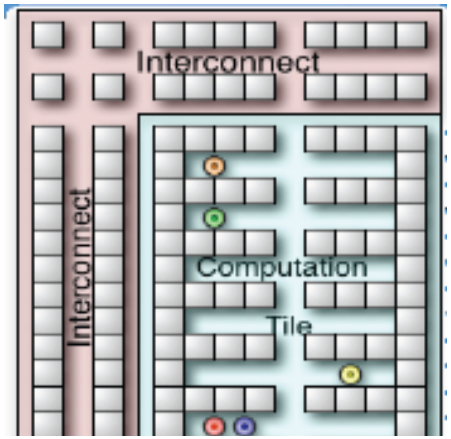
Aliferis, Gottesman, Preskill
FT Proof (2.7 x 10-5)

Aliferis, Cross
FT Proof (1.9 x 10-4)

Knill / Reichardt (1 x10-2)
Threshold estimates
(High overhead ancilla factories)

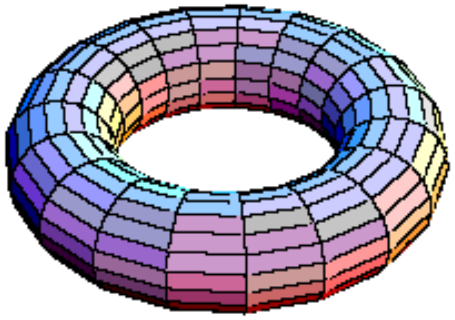
# Developments

Spatial architecture




Gottesman / Aharonov, Ben-Or (1999)
Still a threshold with “local gates”

Denis, Gottesman, Kitaev, Preskill (2001)
Topological quantum memory
Memory threshold of 10-4
(Fast reliable classical control)



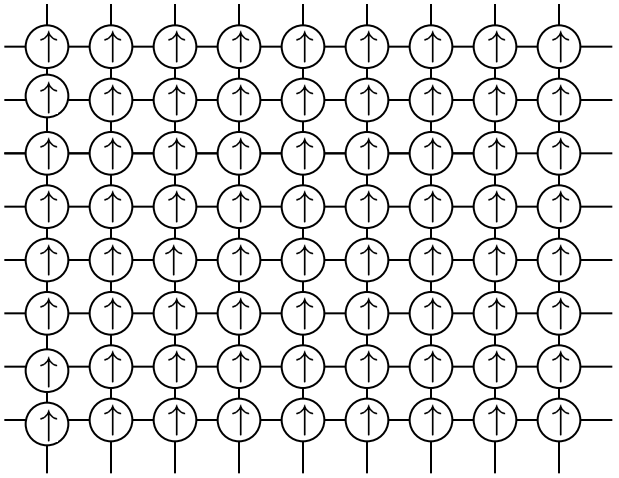
Svore, Terhal, DiVincenzo (2004)
~factor of 10 loss in 2D

Szkopek et al., (2004)
~factor of 100 loss in 1D

Raussendorf, Harrington (2006)
Local in 2D, threshold estimate 5x10-3

# Why Should Physicists Care?

# Toy Classical Model

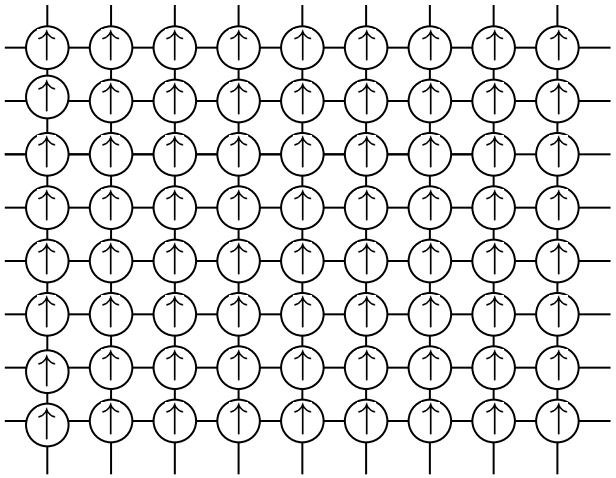


Ising model in two+ dimensions

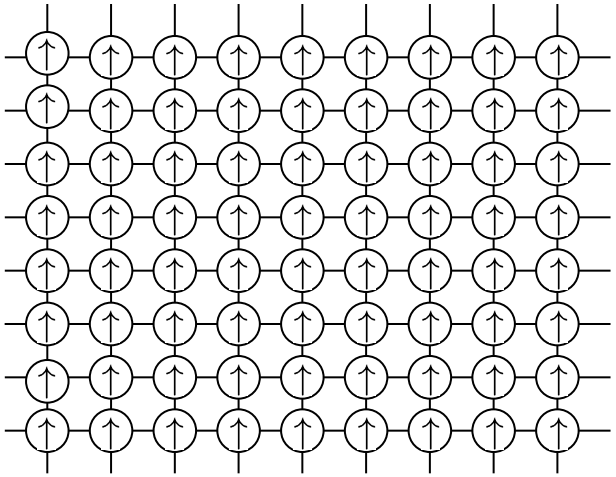
$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

Redundancy code:

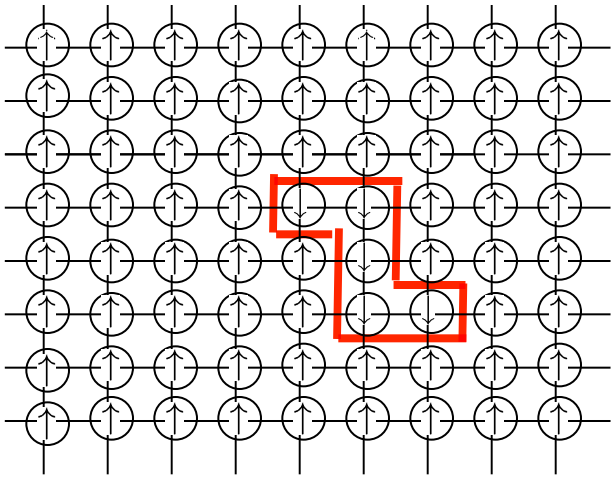
$0 \rightarrow$



$1 \rightarrow$

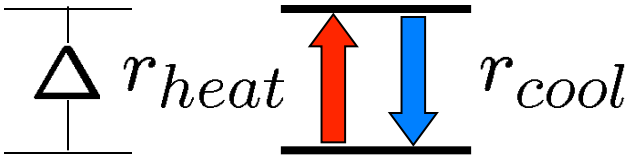


# Dynamical Ising Model



Energy of excitation=
2J (Sum of perimeters of flipped domains)

Dynamics obeying principle of detailed balance:




$$\frac{r_{heat}}{r_{cool}} = e^{-\frac{\Delta}{k_B T}}$$

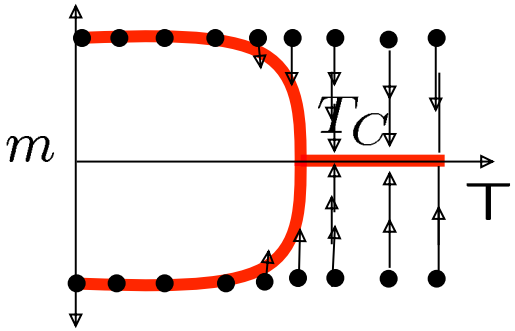
Competition between growth and decimation of flipped domains



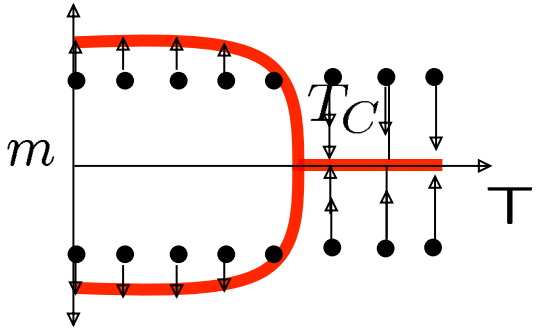
# Robust



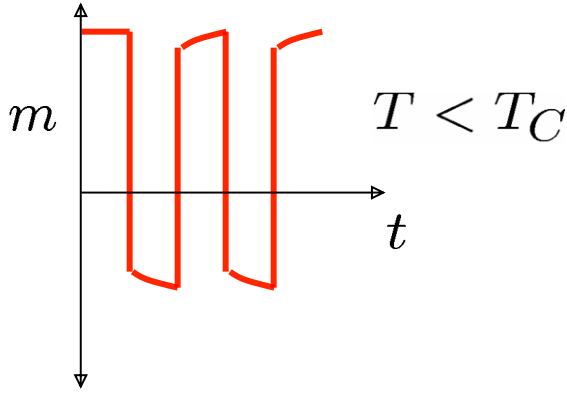
Relaxation to thermal



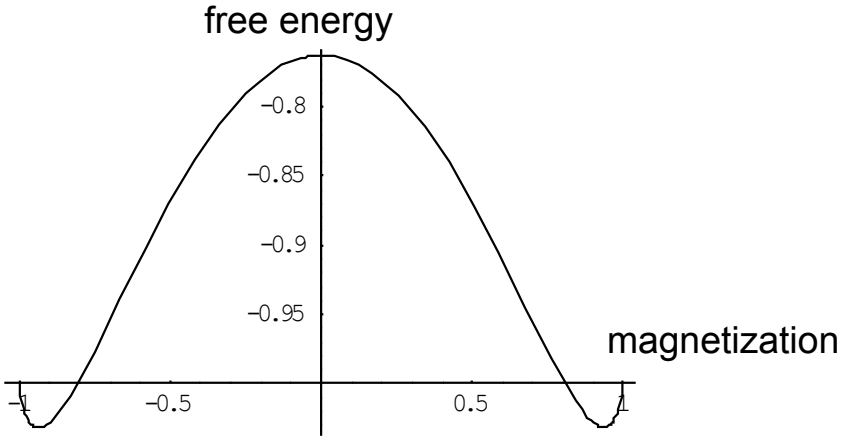

Imperfect preparation




Manipulation

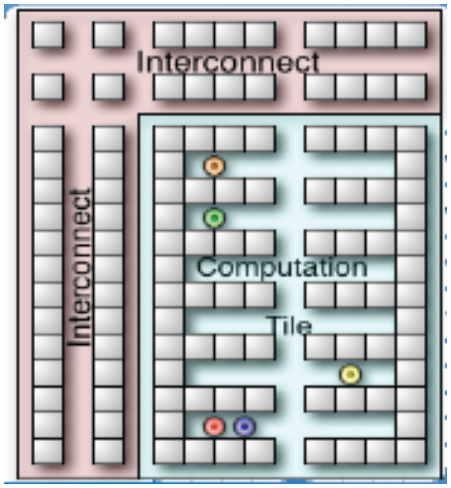



The two encoded states have lifetimes which grow with the size of the system:

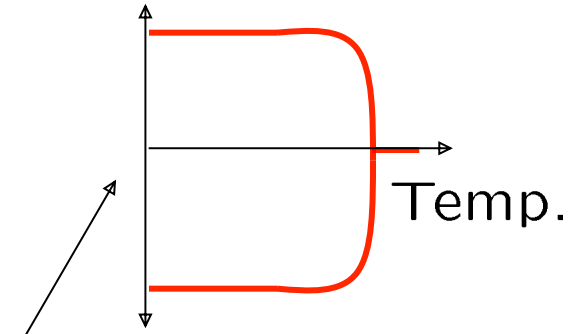


# Why Should Physicists Care?

Spatial architecture




Local quantum many-body system
with a new interesting order parameter:

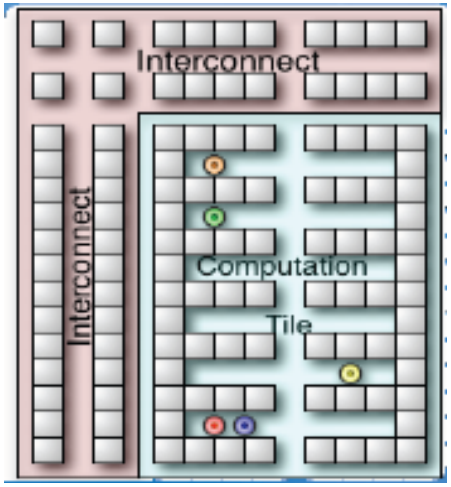



The Quantum Error Correcting Order Parameter(s)

Can we engineer simple many-body quantum systems where quantum error correction is enacted by the physics of the device?

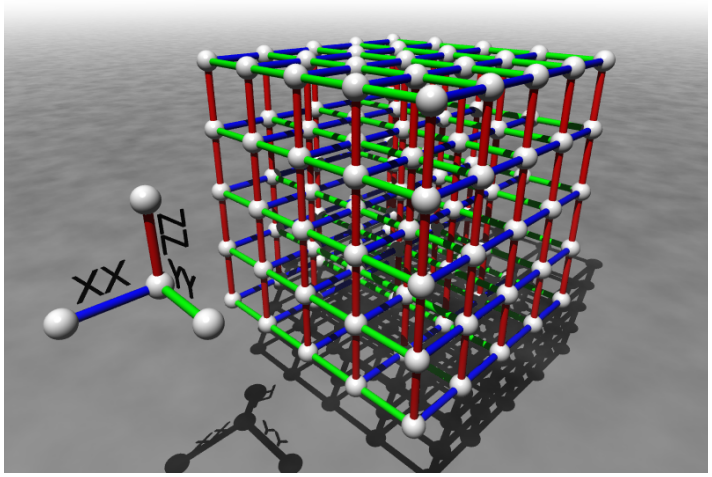
Today: Partial answers to this question. But a grand challenge!

# Our Probable Paths




quantum error correcting architectures

Engineering



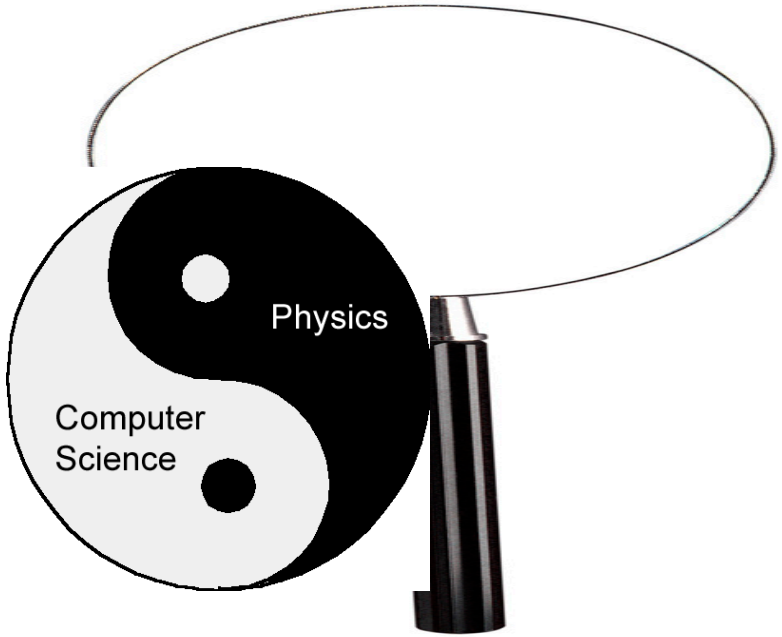

Physics motivated methods for error correction

Physics

# Physics and Computer Science Through the Lens of Quantum Computing

**Dave Bacon**

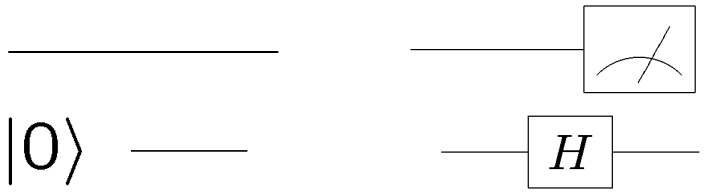
Department
Univers






# Fault-Tolerance

Everything noisy:




Compute on qubits encoded in quantum error correcting code?

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

encoded Paulis

$$\bar{X} = X \otimes X \otimes X \otimes X \otimes X$$
$$\bar{Y} = Y \otimes Y \otimes Y \otimes Y \otimes Y$$
$$\bar{Z} = Z \otimes Z \otimes Z \otimes Z \otimes Z$$

COMPARE

$$\exp\left[-itZ \otimes Z \otimes Z \otimes Z \otimes Z\right] = \cos(t) I - i \sin(t) \bar{Z}$$

$t \approx \frac{\pi}{2}$ Exact: $-i\bar{Z}$ Error: $\bar{Z}$

$$\exp\left[-itZ\right] \otimes \exp\left[-itZ\right] \otimes \exp\left[-itZ\right] \otimes \exp\left[-itZ\right]$$

$t \approx \frac{\pi}{2}$ Exact: $-i\bar{Z}$ Error: single qubit $Z$

$$Z \otimes I \otimes I \otimes I \otimes I$$

# Accuracy?




Error in probabilities is less than

$$E(U,V) = \max_{|||\psi\rangle||^2=1} ||(U-V)|\psi\rangle||$$




Errors <u>add</u> $E(U_m \cdots U_3 U_2 U_1, V_m \cdots V_3 V_2 V_1) \leq \sum_{i=1}^{m} E(U_i, V_i)$

Each gate should be implemented with polynomial accuracy in circuit size

Fault-tolerant universal gate sets:

